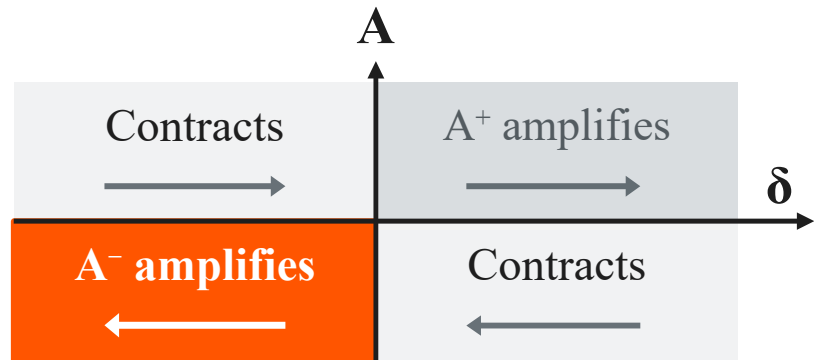


Direction-aware and localized risk



δ : Learner–Sampler mismatch gap

A : Policy advantage

Pre-terminal Stage



Terminal Stage



TRIAGE: selective update control

Diagnose segment S



Gate A^- tokens

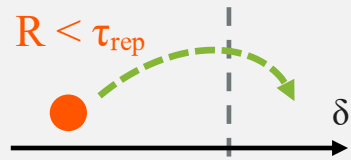


$A^-: w_s$

Rebalance within response

$$\alpha_{i,t} = w_{i,t} / \bar{w}_i$$

Repair severe



$$g_{i,t} = \alpha_{i,t} g_{i,t}^{base} + \psi_{i,t}$$

Performance (Qwen3-30B-A3B)

Rollout throughput (tokens/s)

BF16 3798

TRIAGE 8717 **2.30×**

NVFP4 8942

Average accuracy (%)

Method	Score	Δ vs BF16
BF16	72.41	—
TRIAGE	70.96	-1.45
NVFP4(TIS)	63.18	-9.23